



COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5701

Counting

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Definition of Probability

Let S denote the sample space for the phenomena, the set of all possible outcomes and let the subset E of S be an event.

A function P qualifies as a valid probability measure if and only if it satisfies the following three axioms (Kolmogorov):

1. Axiom of Non-negativity

For any event $E \subset S$, the probability of that event is a non-negative:

$$P(E) \geq 0$$

2. Axiom of Normalization (Unit Measure)

The probability of the entire sample space (the certain event) is exactly 1:

$$P(S) = 1$$

3. Axiom of Countable Additivity (σ -additivity)

For any sequence of mutually exclusive events $E_1, E_2, \dots \subset S$ (where $E_i \cap E_j = \emptyset$ for all $i \neq j$), the probability of their union is equal to the sum of their individual probabilities:

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

The fine print

In rigorous data science contexts, probability is defined over a probability space, represented as a triplet $(\Omega, \mathcal{F}, \mathbb{P})$.

The Probability Space

- **Ω (Sample Space):** The set of all possible outcomes of a random experiment
- **$\mathcal{F}$ (σ -algebra):** A collection of subsets of Ω that represents all strictly measurable events.
- **$\mathbb{P}$ (Probability Measure):** A function that maps events in $\mathcal{F}$ to real numbers, denoted as $\mathbb{P}: \mathcal{F} \rightarrow [0,1]$.

Simple Sample Spaces

In many examples the sample space S contains finite number, N , of outcomes. That is, $S = \{o_1, o_2, o_3, \dots, o_N\}$.

Also, assume each of the outcomes is equally likely. Then

$$P(\{o_i\}) = \frac{1}{N}$$

Such sample spaces are known as **simple** or **equally likely**.

Then for an event E which contains m outcomes we have:

$$P(E) = \frac{m}{N} = \frac{n(E)}{n(S)} = \frac{n(E)}{N} = \frac{\text{\# of outcomes in } E}{\text{total \# of outcomes}}$$

Notes:

- $n(A)$ means the number of elements of A
- The above definition satisfies the axioms of probability!

Example from last time:

Back to example with cars:

Three cars at an intersection can turn either left (L) or right (R).

$A = \text{“Exactly one car turns right”} = \{LLR, LRL, RLL\}$

$C = \text{“All cars turn in the same direction”} = \{LLL, RRR\}$

Find $P(A)$, $P(C)$

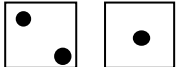
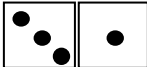
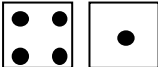
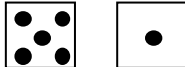
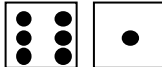
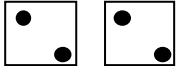
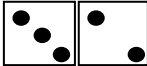
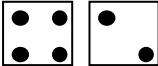
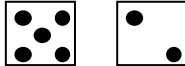
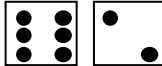
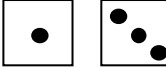
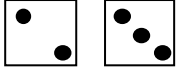
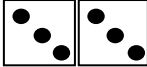
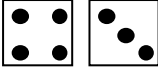
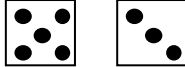
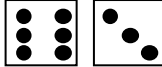
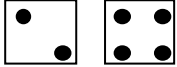
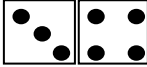
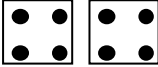
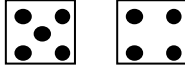
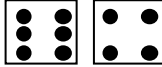
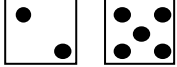
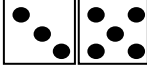
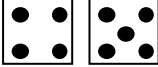
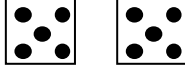
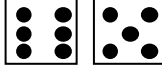
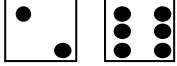
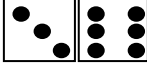
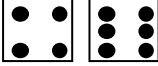
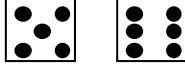
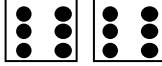
Solution:

$$P(A) = \frac{3}{8} = 0.375$$

$$P(C) = \frac{2}{8} = \frac{1}{4} = 0.25$$

Exercise

Roll two fair dice

Let A = “roll sum greater than or equal to 10”

Find $P(A)$ =

We compute probabilities using rules:

1. $P(E) \geq 0$, for each E .
2. $P(S) = 1$.
3. If A and B are disjoint, then $P(A \cup B) = P(A) + P(B)$
4. $P(\bar{A}) = 1 - P(A)$
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

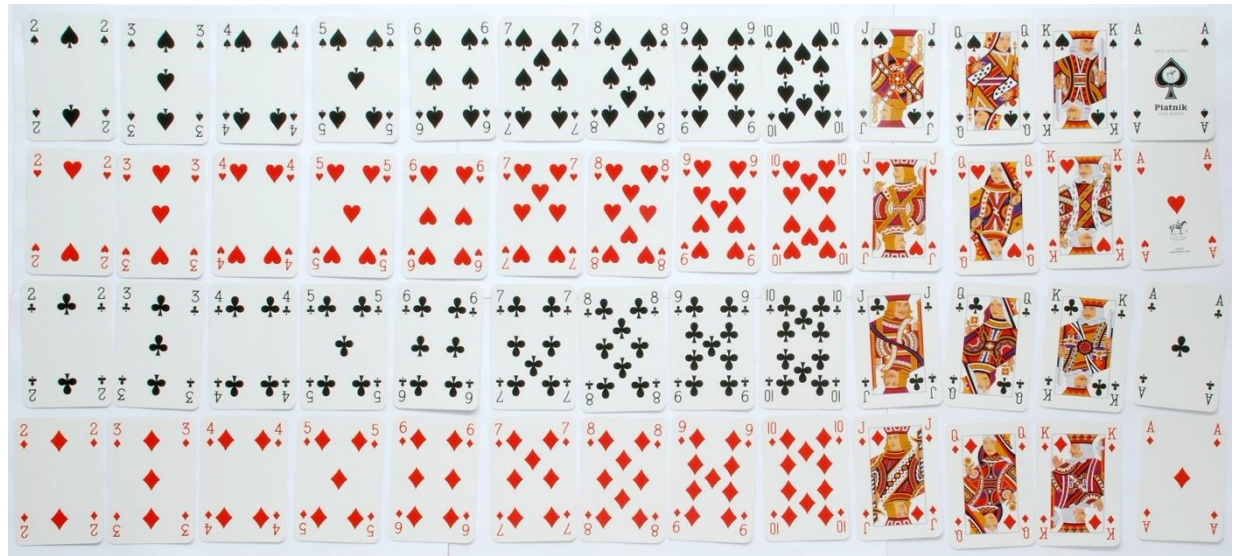
There are more rules. This is the material for next week.

Example 1

In a standard deck of 52 cards, draw one card.

Let W = “ace **or** diamonds”

Find $P(W)$



$$\begin{aligned}\text{Solution: } P(W) &= P(\text{Ace} \cup \text{Diamonds}) \\ &= P(\text{Ace}) + P(\text{Diamonds}) - P(\text{Ace of Diamonds}) \\ &= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}\end{aligned}$$

Counting

Rule 1

Suppose we have a collection of sets $A_1, A_2, A_3, \dots$
and that they are mutually exclusive

(i.e. $A_i \cap A_j = \phi$ for all $i \neq j$)

Let $n_i = n(A_i)$ = the number of elements in A_i .

Let $A = A_1 \cup A_2 \cup A_3 \cup \dots$

Then $N = n(A)$ = the number of elements in A
$$= n_1 + n_2 + n_3 + \dots$$

Rule 2

Suppose we have a sequential experiment

Let

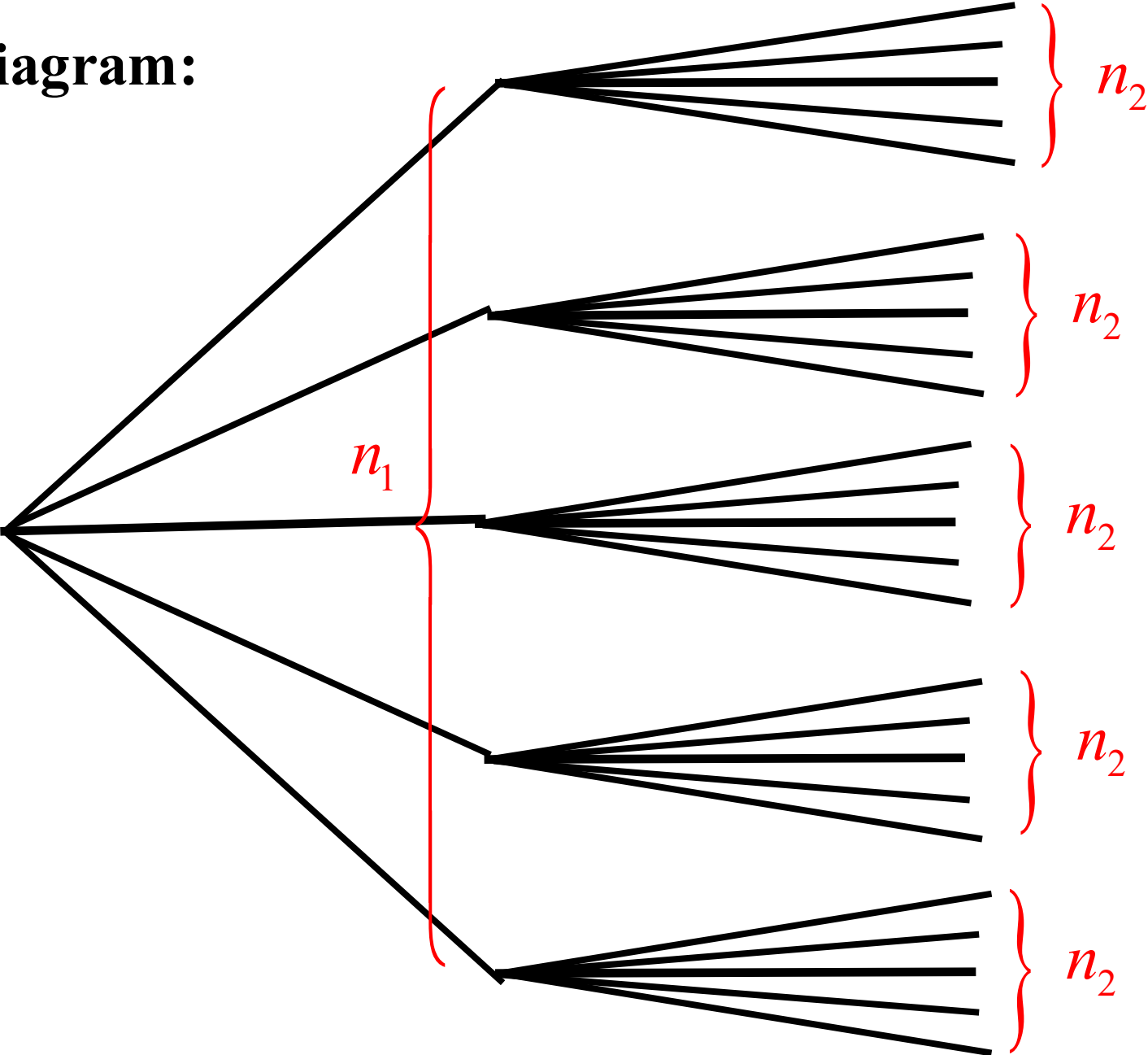
n_1 = the number of ways the first step
can be performed

n_2 = the number of ways the second step
can be performed once the first step
has been completed.

Then $N = n_1 n_2$ = the number of ways the two
operations can be performed in a sequence.

Note: Order matters in sequential experiments!

Diagram:



Example 2

In a championship 100-meter sprint, there are 8 finalists. Assuming there are no ties, in how many ways can the gold and silver medals be awarded?

Step 1: Award the Gold Medal

When the race finishes, any of the 8 runners could potentially cross the finish line first to take the gold.

There are **8 possible choices** for 1st place.

Step 2: Award the Silver Medal

Once the gold medalist is decided, they cannot also win silver. That leaves 7 remaining runners who could cross the finish line second.

There are **7 possible choices** for 2nd place.

Using the fundamental counting principle, we multiply the number of choices for each sequence of events:

$$8 \times 7 = 56$$

Rule 3 (multiplication principle)

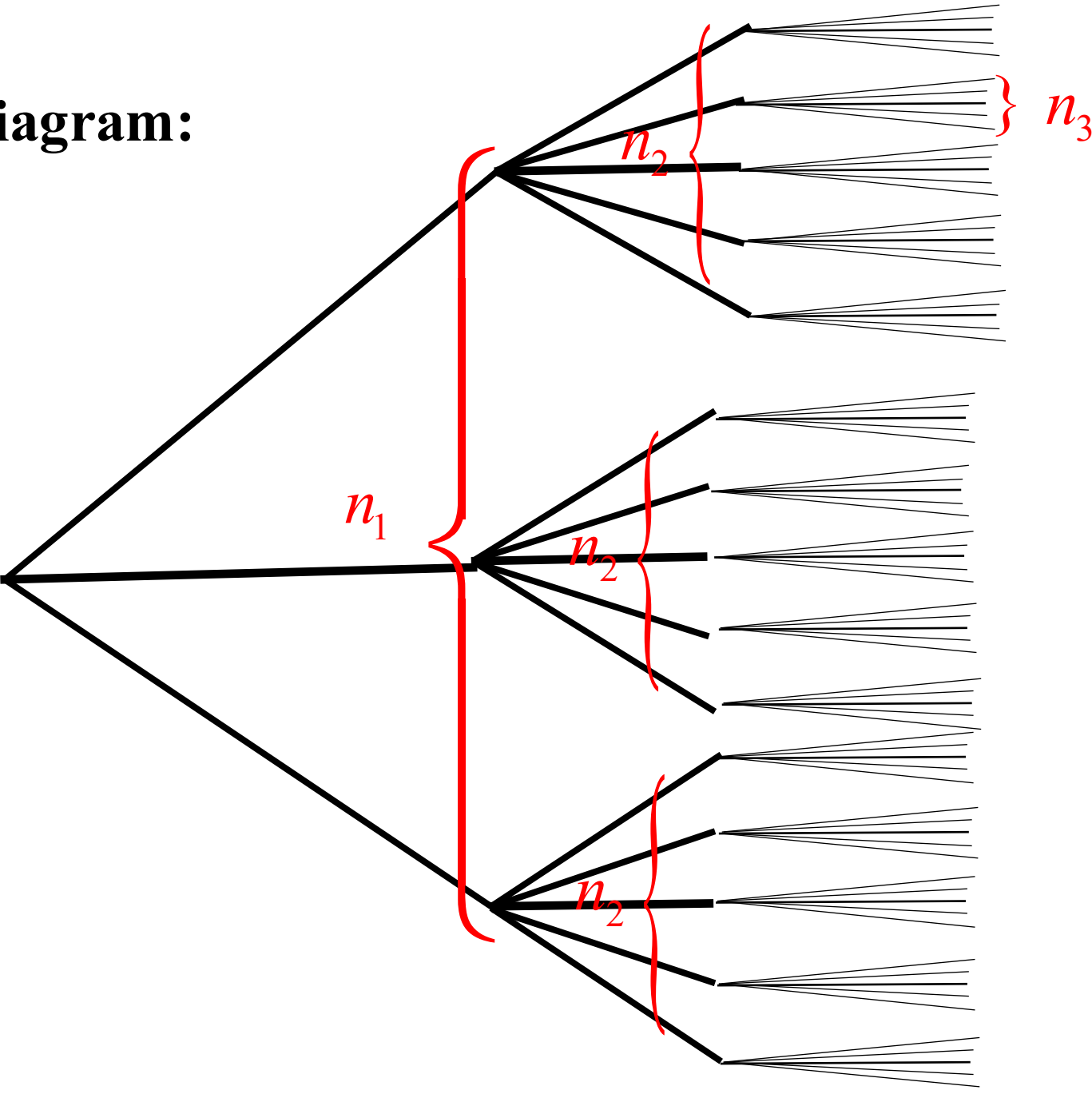
Suppose we carry out k operations in a sequence.

Let:

- n_1 = the number of ways the first operation can be performed
- ...
- n_i = the number of ways the i^{th} operation can be performed once the first $(i - 1)$ operations have been completed, $i = 2, 3, \dots, k$.

Then $N = n_1 n_2 \dots n_k$ = the total number of ways the k operations can be performed in a sequence.

Diagram:



Example 3

How many (theoretically) possible CT license plates could be stamped if each license plate is required to have exactly 2 letters and 5 numbers?

Answer: 67,600,000

Why?



Permutations

In how many ways can you order n objects?

Solution:

Ordering n objects is equivalent to performing n operations in sequence.

1. Choosing the first object in the sequence ($n_1 = n$)
2. Choosing the 2nd object in the sequence ($n_2 = n - 1$).
- ...
- k . Choosing the k^{th} object in the sequence ($n_k = n - k + 1$)
- ...
- n . Choosing the n^{th} object in the sequence ($n_n = 1$)

The total number of ways this can be done is:

$$N = n(n - 1) \dots (n - k + 1) \dots (3)(2)(1) = n!$$

Note: $0! = 1$

Example In how many ways can you order the 4 objects $\{A, B, C, D\}$

Solution:

$$N = 4! = 4(3)(2)(1) = 24$$

Here are the orderings.

<i>ABCD</i>	<i>ABDC</i>	<i>ACBD</i>	<i>ACDB</i>	<i>ADBC</i>	<i>ADCB</i>
<i>BACD</i>	<i>BADC</i>	<i>BCAD</i>	<i>BCDA</i>	<i>BDAC</i>	<i>BDCA</i>
<i>CABD</i>	<i>CADB</i>	<i>CBAD</i>	<i>CBDA</i>	<i>CDAB</i>	<i>CDBA</i>
<i>DABC</i>	<i>DACB</i>	<i>DBAC</i>	<i>DBCA</i>	<i>DCAB</i>	<i>DCBA</i>

Exercise 3:

Mr. Jones has 10 books that he is going to put on his bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Jones wants to arrange his books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

Permutations of size k ($< n$)

In how many ways can you choose k objects from n objects in a specific order

Solution: This operation is equivalent to performing k operations in sequence.

1. Choosing the first object in the sequence ($n_1 = n$)
2. Choosing the 2nd object in the sequence ($n_2 = n - 1$).
- ...
- k . Choosing the k^{th} object in the sequence ($n_k = n - k + 1$)

The total number of ways this can be done is:

$$N = n(n - 1) \dots (n - k + 1) = n! / (n - k)!$$

This number is denoted by the symbol

$$P_{n,k} = {}_n P_k = n(n - 1) \dots (n - k + 1) = \frac{n!}{(n - k)!}$$

Example

In a committee of $n = 10$ people, we want to choose a **chairperson**, a **vice-chairperson** and a **treasurer**.

- **Solution:**
- Essentially, we want to select 3 people from the committee of 10 in a specific order. (Permutations of size 3 from a group of 10).
- $$P(n, k) = P_{n,k} = P_{10,3} = \frac{10!}{(10-3)!} = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$$

Exercise:

- A data scientist is building an ordered 3-stage feature engineering pipeline for a credit risk model by selecting sequentially from a pool of 7 candidate features: {A, B, C, D, E, F, G}.
- The selection must satisfy two domain constraints:
- **Primary Feature Requirement:** The first feature selected (Stage 1) must be a baseline financial metric, which can only be feature A or feature B.
- **Multicollinearity Constraint:** Features C and D are highly correlated, so they cannot both be present in the same 3-stage pipeline.

Important Notes

In **permutations** order is **relevant**. Different orderings result in the different permutations.

If you need to count in a scenario when ordering is **irrelevant**, then you should use **combinations**. In combinations different orderings result in the same combination.

Exercise:

We have a committee of $n = 10$ people, and we want to choose a **chairperson**, a **vice-chairperson** and a **treasurer**. Suppose that 6 of the members of the committee are male and 4 of the members are female. What is the probability that the three executives selected are all male?

Combinations

3. **Combinations of size k ($\leq n$):** A combination of size k chosen from n objects is a subset of size k where the order of selection is irrelevant. How many ways can you choose a combination of size k objects from n objects (order of selection is irrelevant)

Here are the combinations of size 3 selected from the 5 objects $\{A, B, C, D, E\}$

$\{A, B, C\}$	$\{A, B, D\}$	$\{A, B, E\}$	$\{A, C, D\}$	$\{A, C, E\}$
$\{A, D, E\}$	$\{B, C, D\}$	$\{B, C, E\}$	$\{B, D, E\}$	$\{C, D, E\}$

How many ways can you choose a combination $\binom{n}{k}$ of size k objects from n objects (order is irrelevant)?

Solution:

First, the number of ways to pick k objects sequentially when order *does* matter is $P(n, k) = \frac{n!}{(n-k)!}$

In a combination, order does not matter. Any selected group of k distinct objects can be rearranged among themselves in $k!$ different orders:

$$k! = k \times (k - 1) \times \cdots \times 1$$

Since $P(n, k)$ counts each unique set of k objects $k!$ times, the relationship between permutations and combinations $\binom{n}{k}$ is:

$$P(n, k) = \binom{n}{k} \times k!$$

Dividing both sides by $k!$ yields the final combination formula:

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k! (n - k)!}$$

The number:

$$\begin{aligned}\binom{n}{k} &= C(n, k) = \frac{n!}{k! (n - k)!} \\ &= \frac{n(n - 1)(n - 2) \dots (n - k + 1)}{k(k - 1)(k - 2) \dots (1)}\end{aligned}$$

is read “ n choose k ”

- It is the number of ways of choosing k objects from n objects (order of selection irrelevant).
- ${}_nC_k = \binom{n}{k} = C(n, k)$ is also called a **binomial coefficient**.
- It arises when we expand $(x + y)^n$ (**the binomial theorem**)

The Binomial theorem:

$$(x + y)^n = {}_nC_0 x^0 y^n + {}_nC_1 x^1 y^{n-1} + {}_nC_2 x^2 y^{n-2} +$$

$$\text{K} + {}_nC_k x^k y^{n-k} + \text{K} + {}_nC_n x^n y^0$$

$$= \binom{n}{0} x^0 y^n + \binom{n}{1} x^1 y^{n-1} + \binom{n}{2} x^2 y^{n-2} +$$

$$\text{K} + \binom{n}{k} x^k y^{n-k} + \text{K} + \binom{n}{n} x^n y^0$$

On the side: What if n is not a natural number?

Define

$$\binom{\alpha}{k} = \frac{\alpha(\alpha - 1)(\alpha - 2) \dots (\alpha - k + 1)}{k!}, k \in \mathbb{N}, \alpha \in \mathbb{R}$$

Then

$$(1 + x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k, |x| < 1$$

Summary of counting results

Rule 1

$$n(A_1 \cup A_2 \cup A_3 \cup \dots) = n(A_1) + n(A_2) + n(A_3) + \dots$$

if the sets $A_1, A_2, A_3, \dots$ are pairwise mutually exclusive

(i.e. $A_i \cap A_j = \phi$)

Rule 2

$N = n_1 n_2$ = the number of ways that two operations can be performed in sequence if

n_1 = the number of ways the first operation can be performed

n_2 = the number of ways the second operation can be performed once the first operation has been completed.

Rule 3 $N = n_1 n_2 \dots n_k$
 = the number of ways the k operations can be
 performed in sequence if

n_1 = the number of ways the first operation can be
 performed

n_i = the number of ways the i^{th} operation can be
 performed once the first $(i - 1)$ operations have
 been completed. $i = 2, 3, \dots, k$

Summary of basic counting formulas

Permutations (Order Matters)

1. Permutations without Replacement (Standard)

Choosing r items from a pool of n distinct items, where you cannot pick the same item twice.

$$P(n, r) = \frac{n!}{(n - r)!}$$

2. Permutations with Replacement

Choosing r items from a pool of n distinct items, where you are allowed to pick the same item multiple times.

$$n^r$$

3. Permutations of a Multiset (Distinguishable Permutations)

Arranging n total items where some items are identical to each other.

$$\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$$

Summary of basic counting formulas

Combinations (Order Doesn't Matter)

1. Combinations without Replacement (Standard)

Choosing r items from a pool of n distinct items, where you cannot pick the same item twice, often read as " n choose r ."

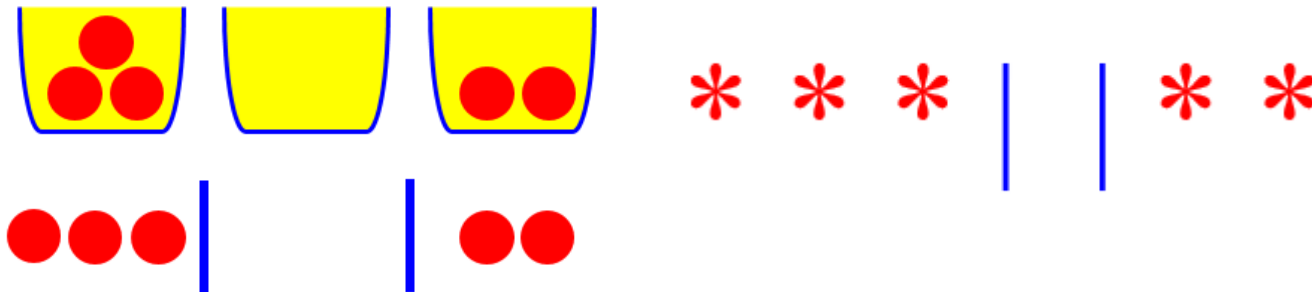
$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

2. Combinations with Replacement (Stars and Bars)

Choosing r items from n distinct categories, where you can pick multiple items from the same category.

Formula: $\binom{n+r-1}{r}$

Example: Buying 5 donuts from a shop that sells 3 different flavors.



Applications to some counting problems

- The trick is to use the basic counting formulae together with the **Rules**
- We will illustrate this with examples
- Counting problems are not easy. The more practice better the techniques

Application to Lotto 6/49

Here you choose 6 numbers from the integers 1, 2, 3, ..., 47, 48, 49.

Six **winning** numbers are chosen.

How many choices for the 6 winning numbers

$$\begin{aligned}\binom{49}{6} &= {}_{49}C_6 = \frac{49!}{6!43!} = \frac{49(48)(47)(46)(45)(44)}{6(5)(4)(3)(2)(1)} \\ &= 13,983,816\end{aligned}$$

You can lose and win in several ways

1. No winning numbers – lose
2. One winning number – lose
3. Two winning numbers - lose
4. Three winning numbers – win \$10
5. Four winning numbers – win approx. \$80
6. 5 winning numbers – win approx. \$2,500
7. 6 winning numbers – win approx. \$4,000,000

Counting the possibilities

1. No winning numbers:

To calculate this, we divide the number of ways to choose only losing numbers by the total number of possible combinations.

Total possible combinations: Choosing 6 numbers out of the total 49.

$$\binom{49}{6} = 13,983,816$$

Ways to choose zero winning numbers: Choosing all 6 of your numbers from the 43 losing numbers (49 total numbers - 6 winning numbers).

$$\binom{43}{6} = 6,096,454$$

Probability:

$$P(0 \text{ matches}) = \frac{6,096,454}{13,983,816} \approx 0.43596$$



1	2	3	4	5	6	7
8	X	10	11	12	13	14
15	16	17	18	X	20	21
22	X	24	25	26	27	28
29	30	31	X	33	34	35
36	37	X	39	40	41	42
43	44	45	46	47	X	49

Counting the possibilities

2. One winning number:

One number is chosen from the six winning numbers and the remaining **five** must be chosen from the losing numbers

$$\binom{6}{1} \binom{43}{5} = 6(962,598) = 5,775,588$$

3. Two winning numbers:

Two numbers are chosen from the six winning numbers and the remaining **four** must be chosen from the losing numbers:

$$\binom{6}{2} \times \binom{43}{4} = 15 \times 123,410 = 1,851,150$$

4. Three winning numbers

Three numbers are chosen from the six winning numbers and the remaining **three** must be chosen from the losing numbers:

$$\binom{6}{3} \binom{43}{3} = 20 \times 12341 = 246,820$$

5. Four winning numbers

Four numbers are chosen from the six winning numbers and the remaining **two** must be chosen from the losing numbers:

$$\binom{6}{4} \binom{43}{2} = 15 \times 903 = 13,545$$

7. five winning numbers

Five numbers are chosen from the six winning numbers and the remaining **number** has to be chosen from the losing numbers (excluding the bonus number)

$$\binom{6}{5} \times \binom{43}{1} = 6 \times 43 = 258$$

8. six winning numbers

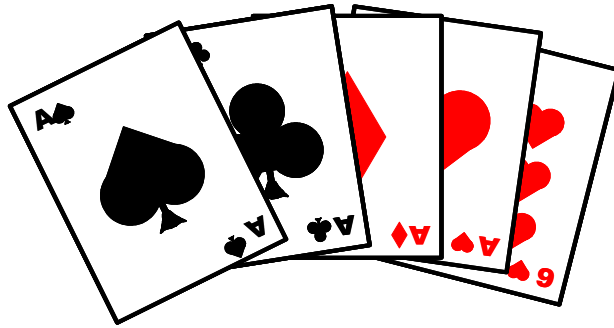
Six numbers are chosen from the six winning numbers,

$$\binom{6}{6} = 1$$

Another Example

counting poker hands

A **poker hand** consists of five cards chosen at random from a deck of 52 cards.



The total number of poker hands is

$$N = \binom{52}{5} = 2,598,960$$

Poker Hands

1. High card $\{x, y, z, u, v\}$
2. Pair $\{x, x, y, z, u\}$
3. Two pairs $\{x, x, y, y, z\}$
4. Three of a kind $\{x, x, x, y, z\}$
5. Straight $\{x, x+1, x+2, x+3, x+4\}$
 - 5 cards in sequence, not all the same suit
6. Flush $\{x, y, z, u, v\}$
 - Not all in sequence but **all the same suit**

POKER HANDS (Ace can be either lowest or highest ranking card)					
A	K	Q	J	10	Royal Flush Highest-ranking straight flush
5	4	3	2	A	Straight Flush 5 cards in rank order, same suit
4	4	4	4		Four of a Kind 4 same rank
3	3	3	2	2	Full House 3 same rank + 2 same rank
♥	♥	♥	♥	♥	Flush 5 cards same suit
5	4	3	2	A	Straight 5 cards in rank order
3	3	3			Three of a Kind 3 same rank
A	A	2	2		Two Pair 2 same rank, twice
2	2				Pair 2 same rank
A					High Card Highest ranking card, no hands

7. Full House $\{x, x, x, y, y\}$



8. Four of a kind $\{x, x, x, x, y\}$



9. Straight Flush $\{x, x + 1, x + 2, x + 3, x + 4\}$

- 5 cards in sequence but not $\{10, J, Q, K, A\}$
- **all the same suit**



10. Royal Flush $\{10, J, Q, K, A\}$

- **all the same suit**



Counting the hands

VISUALIZING ONE PAIR $\{x, x, y, z, u\}$

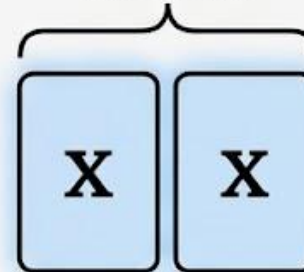
THE HAND

THE PAIR
 x, x



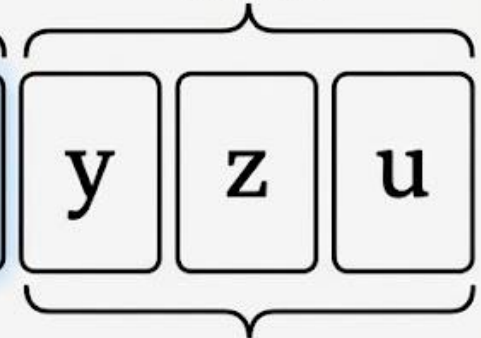
KICKERS
 y, z, u

**Two Matching
Ranks**
(The Pair)



E.g., Two Queens.
There are 13
possible ranks.

**Three Different,
Unique Ranks**
(Kickers)



(Kickers)

E.g., 7, 3, 10.
These must be
distinct ranks.

PROBABILITY CONTEXT



Commonness:
Approx. 42.26% of
all 5-card hands



Counting the hands

2. Pair $\{x, x, y, z, u\}$: meaning you need exactly one pair of a specific rank (x), and three other cards of completely different ranks (y, z , and u) to ensure you don't accidentally get two pair, three of a kind, or a full house.

To find the favorable outcomes, we break the hand down into the pair and the three kickers:

Choose the rank for the pair (x): There are 13 possible ranks (2 through Ace).

$$\binom{13}{1} = 13$$

Choose the suits for the pair: We need 2 out of the 4 available suits for that rank.

$$\binom{4}{2} = 6$$

Choose the ranks for the three remaining cards (y, z, u): We must choose 3 distinct ranks from the remaining 12 ranks, so they don't pair up with each other or the original pair.

$$\binom{12}{3} = 220$$

Choose the suits for the three remaining cards: Each of the 3 cards can be any of the 4 suits.

$$\binom{4}{1}^3 = 4^3 = 64$$

Multiplying these together gives the total number of ways to draw exactly one pair:

$$13 \times 6 \times 220 \times 64 = 1,098,240$$

Counting the hands

3. Two pair $\{x, x, y, y, z\}$

- We have to:
- Choose the values of x, y $\binom{13}{2} = 78$
 - Select the suits for the for x and y . $\binom{4}{2} \times \binom{4}{2} = 36$
 - Choose the denomination z $\binom{11}{1} = 11$
 - Choose the suit for z - 4

$$\text{Total \# of hands of this type} = 78 \times 36 \times 11 \times 4 = 123,552$$

4. Three of a kind $\{x, x, x, y, z\}$

- We have to:
- Choose the value of x $\binom{13}{1} = 13$
 - Select the suits for the for x . $\binom{4}{3} = 4$
 - Choose the denominations $\{y, z\}$ $\binom{12}{2} = 66$
 - Choose the suits for $\{y, z\}$ - $4 \times 4 = 16$

$$\text{Total \# of hands of this type} = 13 \times 4 \times 66 \times 16 = 54,912$$

7. Full House $\{x, x, x, y, y\}$

- We have to:
- Choose the value of x then y ${}_{13}P_2 = 13(12) = 156$
 - Select the suits for the for x . $\binom{4}{3} = 4$
 - Select the suits for the for y . $\binom{4}{2} = 6$

$$\text{Total \# of hands of this type} = 156 \times 4 \times 6 = 3,696$$

8. Four of a kind $\{x, x, x, x, y\}$

- Choose the value of x in $\binom{13}{1} = 13$ ways
- Select the suits for the x in $\binom{4}{4} = 1$ way
- Choose the value of y in $\binom{12}{1} = 12$ ways
- Choose the suit for y in $\binom{4}{1} = 4$ ways
- Apply multiplication principle:

$$\text{Total \# of hands of this type} = 13 \times 1 \times 12 \times 4 = 624$$

10. Royal Flush $\{10, J, Q, K, A\}$

- **all the same suit**

$$\text{Total \# of hands of this type} = 4 \text{ (no. of suits)}$$

9. Straight Flush $\{x, x+1, x+2, x+3, x+4\}$

- 5 cards in sequence but not $\{10, J, Q, K, A\}$
- **all the same suit**

$$\text{Total \# of hands of this type} = 9 \times 4 = 36 \text{ (no. of suits)}$$

The hand could start with $\{A, 2, 3, 4, 5, 6, 7, 8, 9\}$

5. Straight $\{x, x + 1, x + 2, x + 3, x + 4\}$

- 5 cards in sequence
- Not all the same suit

We have to:

- Choose the starting value of the sequence, x .

Total of 10 possibilities $\{A, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

- Choose the suit for each card

$$4 \times 4 \times 4 \times 4 \times 4 = 1024$$

$$\text{Total \# of hands of this type} = 1024 \times 10 - \underline{36} - 4 = 10200$$



We would have also counted straight flushes and royal flushes that have to be removed

6. Flush $\{x, y, z, u, v\}$

- Not all in sequence but **all the same suit**

We have to:

- Choose the suit 4 choices
- Choose the denominations $\{x, y, z, u, v\}$

$$\binom{13}{5} = 1287$$

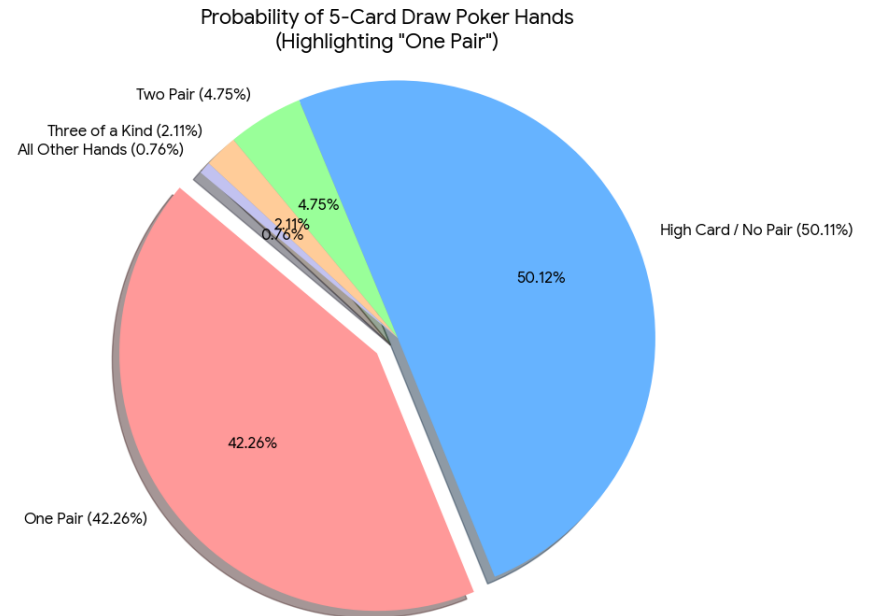
Total # of hands of this type = $1287 \times 4 - \underline{36} - 4 = 5108$



We would have also counted straight flushes and royal flushes that must be removed

Summary

	Frequency	Prob.
nothing	1,302,588	0.50119586
pair	1,098,240	0.42256903
two pair	123,552	0.04753902
3 of a kind	54,912	0.02112845
straight	10,200	0.00392465
flush	5,108	0.00196540
full house	3,696	0.00142211
4 of a kind	624	0.00024010
straight flush	36	0.00001385
royal flush	4	0.00000154
Total	2,598,960	1.00000000



Example

Suppose that a fair coin is to be tossed 10 times, and it is desired to determine the probability of obtaining:

- a) exactly 3 heads
- b) three or fewer heads.

Tennis Tournament Example

Suppose that n tennis players are entered in a tournament. In the first round, the players are paired one against another at random. The loser in each pair is eliminated from the tournament, and the winner in each pair continues into the second round. If the number of players n is odd, then one player is chosen at random before the pairings are made for the first round, and that player automatically continues into the second round. All the players in the second round are then paired at random. Again, the loser in each pair is eliminated, and the winner in each pair continues into the third round. If the number of players in the second round is odd, then one of these players is chosen at random before the others are paired, and that player automatically continues into the third round. The tournament continues in this way until only two players remain in the final round. They then play against each other, and the winner of this match is the winner of the tournament. We shall assume that all n players have equal ability. Find the probability that two specific players A and B will ever play against each other during the tournament.